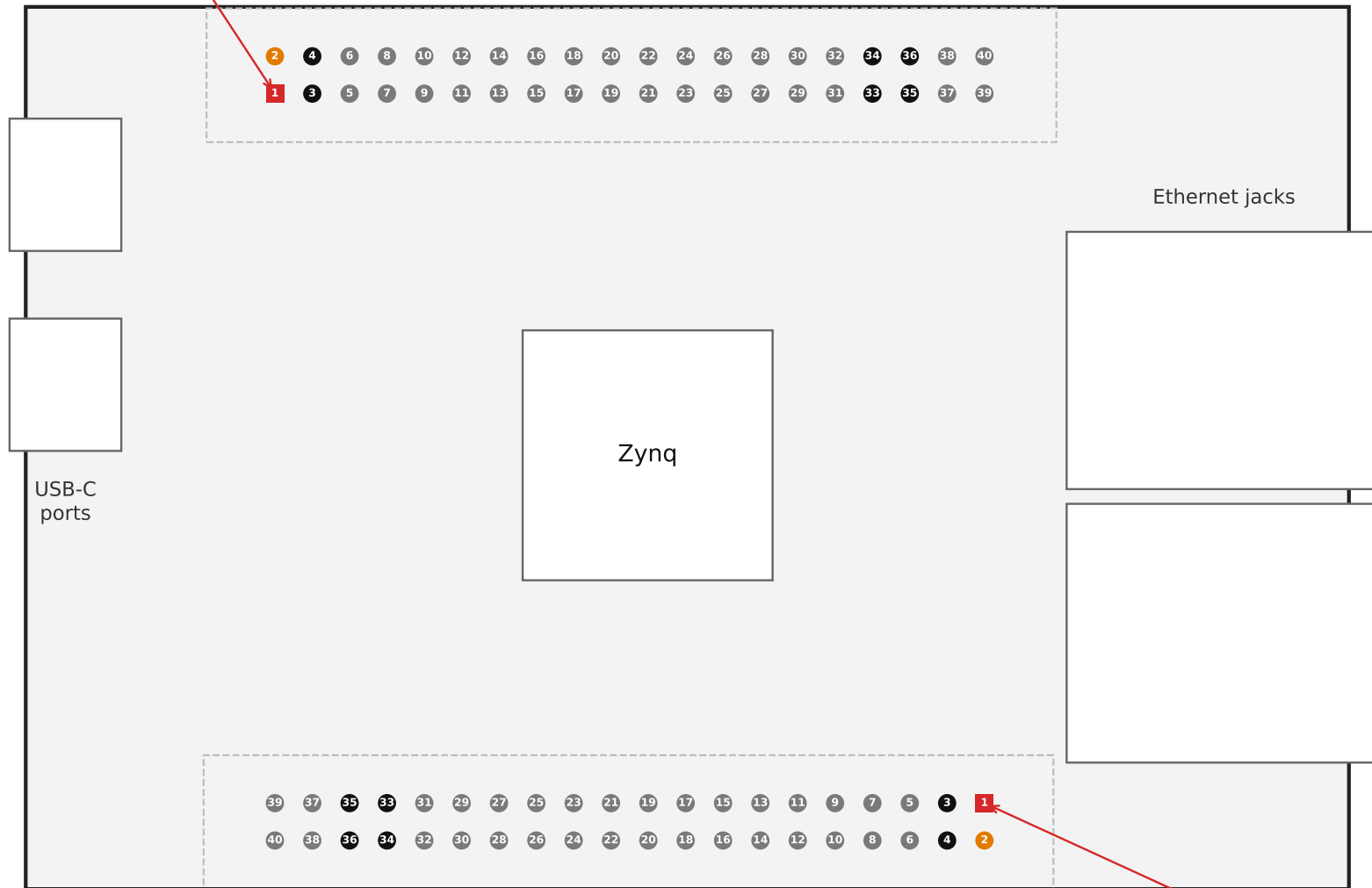


PZ7020-StarLite - 40-pin headers, physical layout (top view, chip side up)

Geometry from Puzhi PZ-STARLITE-PCB-V1_1_TOP.dxf. Hold the board with the two Ethernet jacks on your RIGHT and the two USB-C ports on your LEFT.

pin 1 (5 V) = square pad, LEFT end, inner row

top-edge header: numbering runs left to right



pin 1 (5 V) = square pad, RIGHT end, inner row

■ 5 V rail (in or out)

■ 3.3 V regulator output

■ GND

■ FPGA I/O

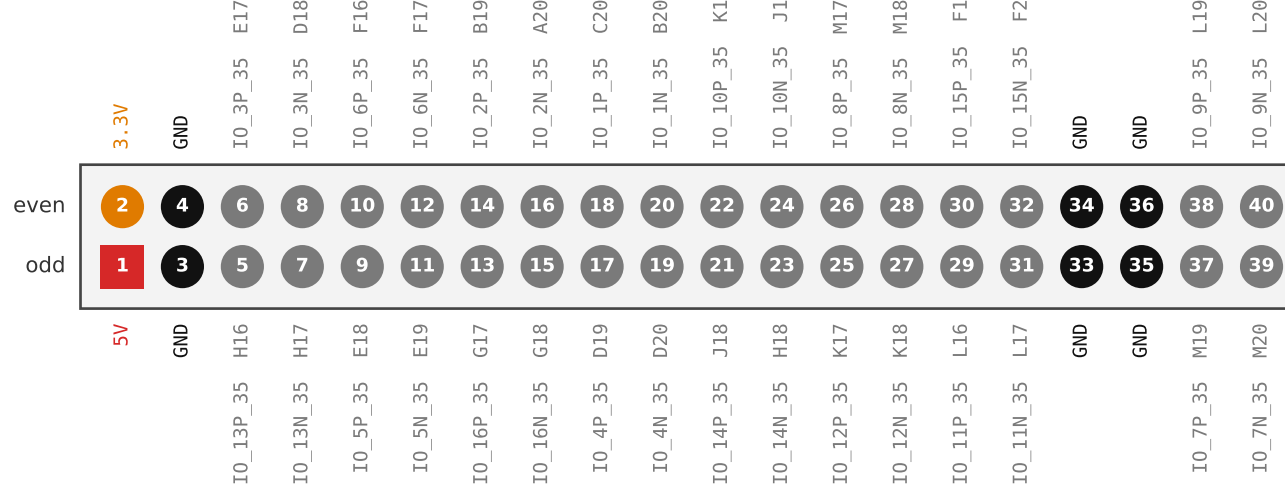
Power pins are IDENTICAL on both headers: 1 = 5 V, 2 = 3.3 V, 3 / 4 / 33 / 34 / 35 / 36 = GND.

Which physical header is JM1 and which is JM2 is NOT stated in any document and is pending a probe test (measurement P-11) - it only matters for the I/O pins.

JM1 - all 40 pins

All 32 I/O on BANK 35 (HR bank, 1.8 / 2.5 / 3.3 V selectable, DEFAULT 3.3 V).
Source: Puzhi pinout xlsx, cross-checked pin-for-pin against schematic sheet 18.

canonical view: pin 1 (square) at the LEFT, numbering to the right, odd row below / even row above - see page 1 for where pin 1 sits on the real board



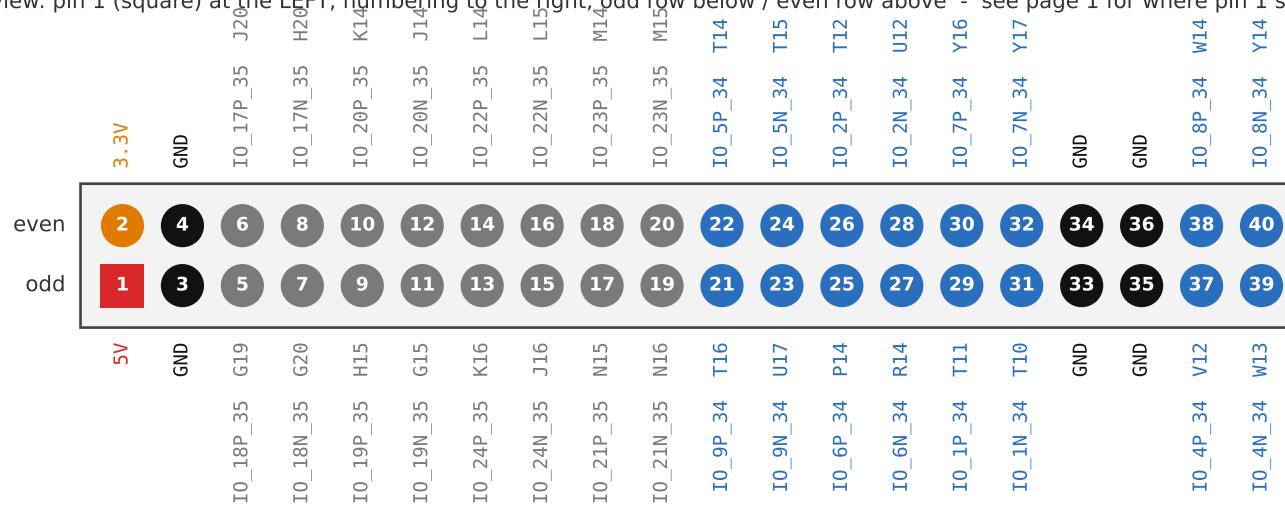
Pin	Net	Ball	Bank / role	Pin	Net	Ball	Bank / role
1	5V	-	+5 V rail (in or out)	2	3.3V	-	3.3 V regulator output
3	GND	-	ground	4	GND	-	ground
5	IO_13P_35	H16	BANK 35	6	IO_3P_35	E17	BANK 35
7	IO_13N_35	H17	BANK 35	8	IO_3N_35	D18	BANK 35
9	IO_5P_35	E18	BANK 35	10	IO_6P_35	F16	BANK 35
11	IO_5N_35	E19	BANK 35	12	IO_6N_35	F17	BANK 35
13	IO_16P_35	G17	BANK 35	14	IO_2P_35	B19	BANK 35
15	IO_16N_35	G18	BANK 35	16	IO_2N_35	A20	BANK 35
17	IO_4P_35	D19	BANK 35	18	IO_1P_35	C20	BANK 35
19	IO_4N_35	D20	BANK 35	20	IO_1N_35	B20	BANK 35
21	IO_14P_35	J18	BANK 35	22	IO_10P_35	K19	BANK 35
23	IO_14N_35	H18	BANK 35	24	IO_10N_35	J19	BANK 35
25	IO_12P_35	K17	BANK 35	26	IO_8P_35	M17	BANK 35
27	IO_12N_35	K18	BANK 35	28	IO_8N_35	M18	BANK 35
29	IO_11P_35	L16	BANK 35	30	IO_15P_35	F19	BANK 35
31	IO_11N_35	L17	BANK 35	32	IO_15N_35	F20	BANK 35
33	GND	-	ground	34	GND	-	ground
35	GND	-	ground	36	GND	-	ground
37	IO_7P_35	M19	BANK 35	38	IO_9P_35	L19	BANK 35
39	IO_7N_35	M20	BANK 35	40	IO_9N_35	L20	BANK 35

JM2 - all 40 pins

Pins 5-20 on BANK 35, pins 21-40 on BANK 34 (blue). BANK 34 also carries HDMI, the user LEDs and the PL reset.

Source: Puzhi pinout xlsx, cross-checked pin-for-pin against schematic sheet 18.

canonical view: pin 1 (square) at the LEFT, numbering to the right, odd row below / even row above - see page 1 for where pin 1 sits on the real board



Pin	Net	Ball	Bank / role	Pin	Net	Ball	Bank / role
1	5V	-	+5 V rail (in or out)	2	3.3V	-	3.3 V regulator output
3	GND	-	ground	4	GND	-	ground
5	IO_18P_35	G19	BANK 35	6	IO_17P_35	J20	BANK 35
7	IO_18N_35	G20	BANK 35	8	IO_17N_35	H20	BANK 35
9	IO_19P_35	H15	BANK 35	10	IO_20P_35	K14	BANK 35
11	IO_19N_35	G15	BANK 35	12	IO_20N_35	J14	BANK 35
13	IO_24P_35	K16	BANK 35	14	IO_22P_35	L14	BANK 35
15	IO_24N_35	J16	BANK 35	16	IO_22N_35	L15	BANK 35
17	IO_21P_35	N15	BANK 35	18	IO_23P_35	M14	BANK 35
19	IO_21N_35	N16	BANK 35	20	IO_23N_35	M15	BANK 35
21	IO_9P_34	T16	BANK 34	22	IO_5P_34	T14	BANK 34
23	IO_9N_34	U17	BANK 34	24	IO_5N_34	T15	BANK 34
25	IO_6P_34	P14	BANK 34	26	IO_2P_34	T12	BANK 34
27	IO_6N_34	R14	BANK 34	28	IO_2N_34	U12	BANK 34
29	IO_1P_34	T11	BANK 34	30	IO_7P_34	Y16	BANK 34
31	IO_1N_34	T10	BANK 34	32	IO_7N_34	Y17	BANK 34
33	GND	-	ground	34	GND	-	ground
35	GND	-	ground	36	GND	-	ground
37	IO_4P_34	V12	BANK 34	38	IO_8P_34	W14	BANK 34
39	IO_4N_34	W13	BANK 34	40	IO_8N_34	Y14	BANK 34

Power pins - what they are and what not to do

HOW THE 5 V PIN IS WIRED (schematic sheets 2 "POWER", 18 "40P CON", 19 "USB2JTAG")

- JM1 pin 1, JM2 pin 1 and the JTAG/power Type-C VBUS pins are ALL the same net, VDD_5V.
- VDD_5V feeds the four MP2143 buck converters (1.0 / 1.35-1.5 / 1.8 / 3.3 V) directly.
- There is NO diode, NO ideal-diode controller, NO load switch and NO fuse between any of those sources.
(The only diodes on the whole board are two 1N4148 signal diodes, three UBQ10A05 ESD arrays and LEDs.)

WHAT THAT MEANS

- The header 5 V pin is BOTH an input and an output - it is simply the rail.
 - * Board powered from the Type-C -> 5 V appears on pin 1 of both headers (this is how the fan is fed).
 - * 5 V applied to pin 1 of either header -> the whole board runs (the manual's "use it as a SOM" mode).
- NEVER connect two 5 V sources at the same time (e.g. the JTAG Type-C into a PC AND an external 5 V on pin 1).
With no isolation they are paralleled hard; the higher one back-drives the other - into the PC's USB port or into whatever is feeding the header. The manual's "select one of the two power methods" is this rule.
- The UART Type-C is safe to leave connected with any power source: its VBUS goes only to VDD_USB2UART_5V, which powers the CH340E alone and does not touch VDD_5V (sheet 9). This is also why that port never lights the power LED.
- Pin 2 (3.3 V) is a regulator OUTPUT (U4, MP2143 -> VDD_3V3). Never feed 3.3 V into it. Use it only to power small 3.3 V loads, within what the regulator can spare from the board's own needs.
- Ratings: board supply is 5 V / 1 A (manual). The 3.3 V regulator's spare capacity is not specified - treat it as tens of mA.

POWERING THE FPGA FROM AN ORANGE PI 4 PRO 5 V HEADER PIN

Electrically possible: OPi 5 V pin -> JM pin 1, OPi GND -> JM GND (pin 3). Same 5 V, shared ground. But:

- The Orange Pi 4 Pro ships with a 5 V / 3 A supply. The OPi itself can draw most of that under load; the FPGA needs up to 1 A. Margin is thin -> brown-outs, SD-card init failures, resets. Feasible at idle, not something to rely on.
- The current capacity of the OPi's 40-pin 5 V pins is not documented in the material we have (unverified).
- While the OPi feeds the FPGA, the FPGA's JTAG/power Type-C must NOT be plugged into a PC (second hard-tied source).
To program over JTAG in that setup you need a VBUS-cut (data-only) USB cable - or unplug the OPi 5 V while programming.

Recommended instead: power each board from its own supply and connect ONLY ground + signal wires between them.

If you want one supply for both, use a single 5 V supply of at least 5 A wired in a star to both boards' 5 V inputs - not daisy-chained through the Orange Pi's header.

WHICH PHYSICAL HEADER IS JM1?

Not stated in the manual, schematic, pinout file or PCB drawing, and the board has no silkscreen labels. It does not matter for power (identical pins). For I/O it does: confirm by driving one JM1 ball (e.g. H16) high from a bitstream and probing pin 5 of each header (planned measurement P-11).